RISK MANAGEMENT REVIEW | MARKET RISK

Executive Market Risk Report

2007-2008 Financial Crisis Deep-Dive

Equal-weight portfolio | JPM AIG GS MS AXP C | Jan 2007 - Dec 2008

1 Executive summary

This report tests how standard market risk tools performed on an equally weighted portfolio of six large financial institutions through the 2007-2008 crisis. All figures are reproduced from the accompanying Jupyter notebook. Where the notebook differs from the original report, the difference is stated in Section 9.

DAILY REBALANCED	BUY & HOLD	MAX DRAWDOWN	WORST SINGLE DAY
-74.64%	-70.04%	-82.70%	-20.11%
$10,000 to $2,536	$10,000 to $2,996	low on Nov 20, 2008	Sep 15, 2008 (Lehman)

Key findings

- **Rebalancing did worse than buy & hold in this period.** Daily rebalancing ended 4.60 percentage points lower ($460 less on $10,000). This is one crisis period, so it shows what happened here, not a rule for all downtrends.
- **Normal-distribution risk models understated the tail.** Normal 99% ES was -11.36%, against -17.09% historically.
- **A fat-tailed model fixed VaR but overshot ES.** The fitted Student's t gave a 99% VaR of -14.14% (historical -13.99%) but an ES of -31.86%, nearly double the historical figure.
- **Diversification was weak throughout.** Average pairwise correlation was 0.65 over the period and stayed between about 0.6 and 0.8 in the rolling windows, and all six stocks fell together on Lehman day.
- **Risk was concentrated in AIG.** AIG contributed 22.1% of portfolio volatility but 30.8% of the loss on the worst 1% of days.

BOTTOM LINE

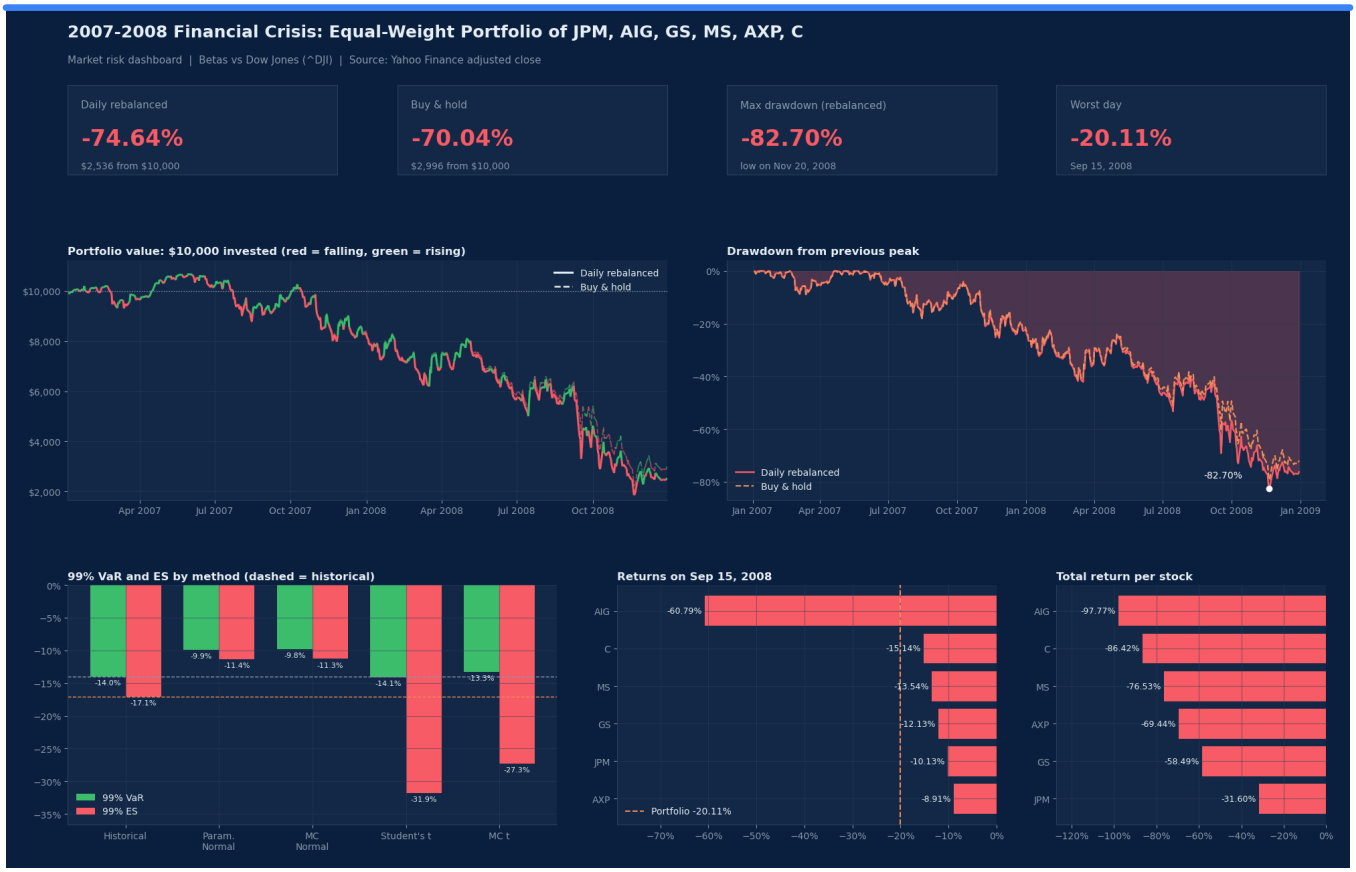
Normal risk models underestimated the crisis, fat-tailed models overshot Expected Shortfall, the six stocks stayed highly correlated throughout, and AIG carried the most tail risk.

2 Data and methodology

- **Data:** daily adjusted closing prices from Yahoo Finance (yfinance), Jan 3, 2007 to Dec 30, 2008 (502 daily returns). The download's end date is exclusive, so Dec 31, 2008 is not included.
- **Portfolio:** equal weights of 1/6 each. The *daily rebalanced* portfolio resets to 1/6 every day. The *buy & hold* portfolio buys equal dollar amounts on day one and never trades. Both start at $10,000.

- **Beta:** covariance of each stock with the benchmark divided by the benchmark's variance. The benchmark is the Dow Jones Industrial Average (^DJI).
- **Volatility:** daily standard deviation multiplied by the square root of 252.
- **Risk measures (99%):** VaR is the loss level exceeded on 1% of days. Expected Shortfall (ES) is the average loss on those days.
- **Five portfolio VaR/ES methods:** historical, parametric Normal, Monte Carlo Normal (10,000 runs, not seeded), parametric Student's t with degrees of freedom fitted by maximum likelihood (scipy), and Monte Carlo from that fitted t-distribution (10,000 runs, seed 42).

Figure 1: Portfolio dashboard



Value line: red where the portfolio fell over the previous 5 days, green where it rose. Solid line is daily rebalanced, dashed is buy & hold.

3 Performance: daily rebalancing vs buy & hold

Portfolio	Final value	Total return
Daily rebalanced	$2,536	-74.64%
Buy & hold	$2,996	-70.04%

How the two differ. With daily rebalancing, each day you sell stocks that did relatively better and buy more of those that did relatively worse, returning every stock to 16.67%. With buy & hold, weights drift: a stock that falls hard becomes a smaller share of the portfolio.

Why rebalancing lost here. AIG (-97.77%) and Citigroup (-86.42%) fell almost without pause, so rebalancing kept adding money to them, while buy & hold let them shrink. Rebalancing tends to help in choppy, mean-reverting markets and hurt in strong one-way trends.

Total return by stock: AIG -97.77%, C -86.42%, MS -76.53%, AXP -69.44%, GS -58.49%, JPM -31.60%.

4 Asset-level risk

Stock	Total return	Annualized volatility	Beta (vs Dow)	99% VaR	99% ES	ES-VaR gap
AIG	**-97.77%**	111%	2.34	**-21.21%**	**-36.17%**	14.96 pts
MS	**-76.53%**	102%	2.56	**-16.88%**	**-22.87%**	5.98 pts
C	**-86.42%**	85%	1.95	**-15.13%**	**-20.93%**	5.80 pts
JPM	**-31.60%**	63%	1.59	**-11.41%**	**-14.55%**	3.14 pts
GS	**-58.49%**	61%	1.51	**-11.04%**	**-13.13%**	2.09 pts
AXP	**-69.44%**	55%	1.59	**-10.45%**	**-13.20%**	2.75 pts

- Volatility of 111% for AIG and 102% for MS means daily swings of several percent were normal.
- Only MS (2.56) and AIG (2.34) had betas above 2. They fell about 2.3 to 2.6 times as much as the Dow on a typical down move.
- The ES-VaR gap measures how much worse the average tail day is than the VaR cutoff. AIG's gap is nearly five times JPM's, which points to much heavier tails. The gap is a difference between two averages, not an extra one-day drop.

5 Portfolio tail risk: VaR and Expected Shortfall (99%)

Method	99% VaR	99% ES	Reading
Historical	**-13.99%**	**-17.09%**	Benchmark. ES rests on about 5 days.
Parametric Normal	**-9.94%**	**-11.36%**	Understates both measures.
Monte Carlo Normal	**-9.77%**	**-11.29%**	Same weakness. Unseeded, so it varies by run.
Student's t (fitted)	**-14.14%**	**-31.86%**	VaR matches history. ES is almost double.
Monte Carlo t (df = 1.80)	**-13.29%**	**-27.35%**	Close to parametric t, same overshoot on ES.

- **Normal models are too optimistic.** They miss about 4 points of VaR and 5.7 points of ES compared with history.
- **The fitted Student's t has df = 1.7989** (location -0.0016, scale 0.0172). A df at or below 2 means the fitted variance is infinite. This shows very heavy tails in this sample. It is a model parameter, not proof of a regime shift.
- **The t-based ES is too large to use as a point estimate.** With df this low, the tail has no stable average, so ES is very sensitive to the fit. It is better read as a conservative upper view.

6 Stress event: Lehman Brothers bankruptcy (Sep 15, 2008)

Stock	AIG	C	MS	GS	JPM	AXP	Portfolio
Return	-60.79%	-15.14%	-13.54%	-12.13%	-10.13%	-8.91%	-20.11%

All six stocks fell by roughly 9% to 61% on the same day across insurance, consumer credit, investment banking and commercial banking. Diversification across these names gave no protection. The portfolio's -20.11% is the average of the six returns, which is exactly what equal weighting implies.

7 Systemic risk and risk concentration (extension)

This section is an addition to the original report. It tests the claim that correlations rise toward 1 in a crisis, and shows where the portfolio's risk came from.

Figure 2: Rolling correlation and risk contribution



Left: average of the 15 pairwise correlations over a rolling 60-day window (first window ends in April 2007). Right: share of portfolio volatility and share of loss on the worst 1% of days, equal weights.

Rolling correlation

- Average pairwise correlation was **0.65** over the full period. The rolling series averaged **0.72** in 2007 and **0.71** in 2008, with a peak of **0.82** on Sep 19, 2007.
- Correlations were already high before the crisis deepened and did not climb steadily toward 1.0 in 2008. The data supports "diversification was weak across these six stocks", not "correlations converged to 1".
- Correlation within a rolling window is itself noisy, and a different window length would give a different picture.

Risk contribution

Stock	Share of portfolio volatility	Share of loss on worst 1% of days	Equal share
AIG	22.1%	30.8%	16.7%
MS	21.5%	19.0%	16.7%
C	18.6%	15.2%	16.7%
GS	13.1%	12.5%	16.7%
JPM	12.9%	11.9%	16.7%
AXP	11.8%	10.6%	16.7%

- AIG and MS together drove 43.6% of volatility and 49.8% of the loss on the worst days, from one-third of the capital.
- AIG's share of the tail loss (30.8%) is well above its share of volatility (22.1%): it mattered most on the worst days.
- The worst-day figures rest on about 5 days, so treat them as a rough indicator. Both measures use fixed 1/6 weights, so they describe the daily rebalanced portfolio.

8 Recommendations

Recommendation	Support in this analysis	Caveat
Avoid mechanical daily rebalancing in a sector downtrend. Consider band-based rebalancing.	Rebalanced finished 4.60 points below buy & hold.	One crisis period, with no test in other markets.
Do not use Normal VaR for capital decisions. Use ES from a fat-tailed model (Student's t or Extreme Value Theory).	Normal VaR and ES were well below history. t-based VaR matched history.	t-based ES overshot history by almost 2x, so it needs calibration and backtesting.
Force correlations to 1.0 in forward stress tests.	Average correlation was 0.65 to 0.82 here, so history already showed little diversification.	Treat 1.0 as a conservative bound, not an expected outcome.
Monitor single-name concentration (added).	AIG supplied 30.8% of worst-day losses on a 16.7% weight.	Based on about 5 tail days.

9 Differences from the original report

Original report	This notebook
Betas measured against the S&P 500.	Betas reproduce only with the Dow Jones (^DJI), so the benchmark is labeled accordingly.
df = 1.79 is "mathematical proof of a regime shift".	It shows very heavy tails in this sample. It is a fitted parameter, not proof.
Multivariate asset-level Monte Carlo (df = 2.01): VaR -2.36%, ES -4.35%.	Not reproduced. The figure is very low, likely from how the t-distribution is scaled when df is near 2 (not verified). It is not evidence about correlation.
Correlations converge toward 1.0 in the crisis.	Rolling correlation was high all along (0.72 in 2007, 0.71 in 2008). No clear rise toward 1.0.
AIG's VaR-ES gap is an extra 15% drop in a single session.	The gap is the difference between two averages, not a one-day drop.
GS "preserved capital".	GS still fell 58.49%.
All betas show firms magnified the market by more than 2x.	Only MS (2.56) and AIG (2.34) exceed 2.
Peak-to-trough period ending Dec 31, 2008.	A fixed calendar window. Data ends Dec 30, 2008.
Monte Carlo Normal: VaR -9.76%, ES -11.06%.	-9.77% and -11.29% in this run. The simulation is unseeded, so values vary.
t and direct Monte Carlo labeled "hyper-conservative" and "robust".	Both match VaR and overshoot ES (-31.86% and -27.35% vs -17.09%), so they are described the same way here.

10 Limitations and model risk

Data

- Prices are Yahoo Finance adjusted closes. Other sources can differ slightly (AIG and JPM total returns differ by 0.01 point from the original report).
- The sample has about 500 trading days. At 99% confidence the tail holds about 5 observations, so historical VaR and ES are noisy.

Portfolio design

- The six firms were chosen with hindsight. Lehman, Bear Stearns, Merrill Lynch and Washington Mutual are not included, so the results do not represent the whole financial sector.
- Equal weights are an assumption. Transaction costs and taxes are ignored, which favors daily rebalancing in the comparison.

Conclusions

- The rebalancing result comes from one crisis period and is not a tested rule.
- The fitted df of 1.80 reflects heavy tails in this sample. Real returns do not have truly infinite variance.

Models

- The Student's t Monte Carlo uses seed 42 and 10,000 runs. The Normal Monte Carlo is not seeded, so its results change each run.
- Normal VaR and ES assume normally distributed returns, which this data clearly rejects.
- No formal VaR backtest (counting breaches) was done, so model accuracy is judged only by comparison with historical values.
- The original report's multivariate Monte Carlo is not included.

A Appendix: definitions

Term	Meaning
Value at Risk (VaR)	The loss level exceeded on only 1% of days at 99% confidence.
Expected Shortfall (ES)	The average loss on the days that exceed VaR. Always worse than VaR.
Beta	How much a stock moves for a 1% move in the benchmark.
Drawdown	The fall from the previous peak in portfolio value.
Fat tails	Extreme moves happen more often than a Normal distribution predicts.
Degrees of freedom (df)	Controls how heavy the Student's t tails are. Lower is heavier. At 2 or below, variance is infinite.
Risk contribution	A stock's share of portfolio volatility, based on its weight and its covariance with the portfolio.